

ELITE MASTER APTITUDE ROADMAP

Phase 0 — Mathematical Foundation (Don't Skip)

These are prerequisites. Many students fail because they jump directly to shortcuts.

Number Basics

Natural Numbers

Whole Numbers

Integers

Rational Numbers

Irrational Numbers

Real Numbers

Complex Numbers (basic idea)

Place Value

Face Value

Number Line

Divisibility

Divisibility Rules (2–20)

Prime Numbers

Composite Numbers

Twin Primes

Co-prime Numbers

Prime Factorization

Fractions

Types

Simplification

Comparison

Operations

Decimals

Conversion

Operations

Recurring decimals

Surds and Indices

Square roots

Cube roots

Laws of exponents

Rational exponents

Approximation

Estimation

Rounding

Significant figures

Phase 1 — Number System (Complete Mastery)

Factors

Multiples

LCM

HCF

Euclid Algorithm

Remainders

Modular Arithmetic

Cyclicity

Last Digit Problems

Unit Digit

Digital Root

Number Properties

Even/Odd

Divisibility

Base Conversion

Binary

Octal

Hexadecimal

Number Sequences

Difficulty:

GATE ★★★

Placements ★★★★★

Government ★★★★★

Phase 2 — Arithmetic (Largest Section)

Percentages

Basic Percentage

Successive Percentage

Reverse Percentage

Population Growth

Income/Expenditure

Percentage Change

Ratio & Proportion

Direct Ratio

Inverse Ratio

Continued Ratio

Compound Ratio

Partnership

Sharing

Average

Weighted Average

Combined Average

Running Average

Replacement Average

Profit and Loss

Discount

Marked Price

Cost Price

Selling Price

False Weight

Successive Discounts

Simple Interest

Compound Interest

Annual

Half Yearly

Quarterly

Growth Problems

Depreciation

Time and Work

Work Efficiency

Alternate Working

Pipes and Cisterns

Man-Days

Work Wages

Time Speed Distance

Relative Speed

Average Speed

Boats

Streams

Trains

Circular Tracks

Race Problems

Mixture & Alligation

Replacement

Mixing

Concentration

Ages

Linear Equations

Family Age Problems

Partnership

Calendar

Clock

Pipes and Cisterns

Race Problems

Boats and Streams

Work and Wages

Installments

Population

Growth and Decay

Simple Equations

Phase 3 — Algebra

Linear Equations

Quadratic Equations

Polynomial

Identities

Inequalities

Absolute Value

Functions

Graphs

Logarithms

Exponents

Progressions

Binomial Expansion (basic)

Coordinate Geometry basics

Phase 4 — Geometry

Lines

Angles

Triangles

Similarity

Congruency

Area

Quadrilaterals

Circles

Polygon

Coordinate Geometry

Geometry Theorems

Phase 5 — Mensuration

2D

Square

Rectangle

Circle

Triangle

Rhombus

Trapezium

Area

Perimeter

3D

Cube

Cuboid

Cylinder

Cone

Sphere

Hemisphere

Frustum

Surface Area

Volume

Phase 6 — Data Interpretation

Tables

Pie Charts

Line Graphs

Bar Graphs

Mixed Graphs

Radar Charts

Caselets

Calculation Speed

Approximation

Phase 7 — Statistics

Mean

Median

Mode

Weighted Mean

Variance

Standard Deviation

Range

Frequency

Histogram

Ogive

Phase 8 — Probability

Sample Space

Events

Conditional Probability

Independent Events

Bayes Theorem

Binomial Probability

Phase 9 — Permutation & Combination

Fundamental Counting Principle

Permutation

Combination

Circular Arrangement

Restricted Arrangement

Distribution Problems

Phase 10 — Logical Reasoning

Series

Number Series

Alphabet Series

Analogies

Classification

Coding Decoding

Blood Relations

Direction Sense

Ranking

Order Arrangement

Seating Arrangement

Linear

Circular

Square

Complex

Puzzles

Input Output

Statement Assumption

Statement Conclusion

Statement Argument

Syllogism

Venn Diagram

Cause Effect

Decision Making

Data Sufficiency

Logical Deductions

Phase 11 — Analytical Reasoning

Logical Connectives

Truth Tables

Deduction

Induction

Critical Reasoning

Pattern Recognition

Numerical Reasoning

Matrix Reasoning

Phase 12 — Spatial Aptitude (Important for GATE)

As specified in the GATE General Aptitude syllabus, this includes transformation of shapes, paper folding, paper cutting, mirroring, assembling, grouping, scaling, rotation, translation, and recognizing 2D/3D patterns. 

GA_2026_Syllabus (1).pdf

Topics:

Mirror Images

Water Images

Paper Folding

Paper Cutting

Cube Problems

Dice

Figure Completion

Embedded Figures

Figure Counting

Rotation

Reflection

Symmetry

Pattern Assembly

Phase 13 — Verbal Ability

Grammar:

Parts of Speech

Tenses

Subject-Verb Agreement

Articles

Prepositions

Conjunctions

Modifiers

Active/Passive Voice

Direct/Indirect Speech

Vocabulary:

Synonyms

Antonyms

Idioms

Phrases

One-word Substitutions

Prefixes/Suffixes

Root Words

Reading:

Reading Comprehension

Para Jumbles

Sentence Rearrangement

Cloze Test

Fill in the Blanks

Sentence Improvement

Error Detection

Critical Reading

Narrative Sequencing

Phase 14 — Exam-Specific Practice

GATE

General Aptitude Previous Year Questions

Timed practice

Mixed-topic tests

NAT, MSQ, MCQ strategy

Placements

Company-wise aptitude (TCS, Infosys, Accenture, Capgemini, Cognizant, Wipro, Deloitte, etc.)

Speed and accuracy drills

Mock aptitude rounds

Government Exams

SSC CGL/CHSL

Banking (IBPS, SBI)

Insurance

Railways

State PSC

Daily sectional tests

Full-length mocks

Phase 15 — Mastery

Formula Revision Notebook

Short Tricks (only after understanding concepts)

Error Log

Weak Topic Analysis

Timed Sectional Tests

Full Mock Tests

Previous Year Papers

Adaptive Practice

Speed Optimization

Accuracy Improvement

Mental Calculation Techniques